

incoming queries.

11.20.3. Software update workflow

The software update workflow consists of several steps executed in a sequence from an OTA Requestor towards an OTA Provider. When a newer Software Image for an OTA Requestor is available on the OTA Provider this results in an updated Software Image being acquired and applied by said OTA Requestor.

The steps, in order, and assuming each step successfully leads to the next, are the following, with each numbered according to [Figure 96, “Detailed OTA Software Update Workflow”](#):

- [10] OTA Provider optionally announces its presence to nodes (see [Section 11.20.7.6.1, “AnnounceOTAProvider Command”](#)). This MAY be used in addition to other OTA Provider discovery methods.
- [11] OTA Requestor determines OTA Provider to query.
- [11] OTA Requestor queries the OTA Provider for availability of an updated Software Image version.
- [30..34] OTA Provider obtains consent from user to apply the OTA update.
- [40..41] OTA Provider obtains a copy of the new Software Image, either in real time or in a time-deferred manner, to provide to the OTA Requestor over [BDX Protocol](#), or over an alternate supported protocol that both OTA Provider and OTA Requestor support. If the Software Image happens to be already available in the OTA Provider’s [cache](#), this step can be skipped.
- [52] OTA Requestor downloads the update, either over [BDX Protocol](#) from OTA Provider acting as a proxy, or over an alternate protocol that both OTA Provider and OTA Requestor support.
- [60] OTA Requestor notifies the OTA Provider that the download is complete and that it is ready to apply the downloaded image.
- [61] OTA Provider responds with an authorization to apply the update, after an optional delay.
- [62] OTA Requestor applies the update and starts executing updated software.
- [63] OTA Requestor notifies the OTA Provider of having successfully applied the update.

In order to illustrate more specifically these steps, [Figure 96, “Detailed OTA Software Update Workflow”](#) below depicts a detailed sequence showing the following illustrative (not normative) aspects:

- [10..21] Determining the availability of an OTA software update.
- [22] Deferral of download by OTA Provider responding with a "Busy" condition, while it obtains user consent and obtains the Software Image from a vendor server based on information in the OTA Provider’s OTA Software Update logic.
- Obtaining user consent in one of these ways:
 - [30..31] Out-of-band notification through some externally-provided user interface, such as a mobile device terminal operated by an authorized user, and connected to OTA Provider’s logic in some way.
 - [32..34] Reuse of prior user consent, perhaps from a continued but revokable authorization, sent back to the OTA Provider by OTA Software Update logic.